NPTEL Week 4

ASSIGNMENT 4 SOLUTIONS

PROGRAMMING SOLUTIONS

1. Max-Min Temperature Difference

Calculates temperature range from inputs. Prevents trailing newlines via end=''.

// PYTHON

```
a = [int(x) for x in input().split()]

if len(a) == 0:
    print(0, end='')
else:
    print(max(a) - min(a), end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `a = [int(x) for x in input().split()]
if len(a) == 0:
    print(0, end='')
else:
    print(max(a) - min(a), end='')`);
}
```

2. Prime Number Checker

Optimized primality test searching factors up to floor(sqrt(n)).

// PYTHON

```
a = int(input())

if a < 2:
    print(False, end='')
else:
    is_prime = True
    for i in range(2, int(a ** 0.5) + 1):
        if a % i == 0:
            is_prime = False
            break
    print(is_prime, end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `a = int(input())

if a < 2:
    print(False, end='')
else:
    is_prime = True
    for i in range(2, int(a ** 0.5) + 1):
        if a % i == 0:
            is_prime = False
            break
    print(is_prime, end='')`);
}
```

3. Duplicate Product IDs

Compares original sequence length against set cardinality.

// PYTHON

```
items = input().split()
print(len(items) != len(set(items)), end='')
```

// CONSOLE INJECTION

```
const el = document.querySelector('textarea.inputarea, textarea');
if (el) {
  el.focus();
  document.execCommand('selectAll', false, null);
  document.execCommand('insertText', false, `items = input().split()
print(len(items) != len(set(items)), end='')`);
}
```

MCQ ANSWERS

Q01 To ensure that February is assigned the correct number of days.

Q02 datetime

Q03 To compare birthdays without considering the birth year.

Q04 To make duplicate values easier to identify.

Q05 The random birthday generation creates a different set of birthdays in every run.

Q06 To generate two cards with exactly one common symbol.

Q07 MSQ

—Removing the common symbol from the symbol pool after placing it.

—Removing every newly assigned symbol from the available symbol pool.

—Randomly choosing only one symbol to be shared between both cards.

Q08 To select a symbol randomly from the available symbol list.

Q09 A five-element list has valid indices only from 0 to 4.

Q10 It asks the player to identify the common symbol.

Q11 Nested list comprehension

Q12 The number 1 will always be placed in the middle row of the last column.

Q13 To allow the program to identify cells that have not yet been assigned a value.

Q14 It places only the first number and does not determine positions for the remaining numbers.

Q15 MSQ

—It allocates memory for an n × n matrix before placing any numbers.

—It determines the initial position of the first number using the Magic Square algorithm.

—It uses nested iteration while creating the matrix.

Q16 create_question()

Q17 The current masked title remains unchanged, and a message is displayed.

Q18 To reveal only the newly guessed letters while preserving previously revealed letters.

Q19 All letters in the movie title have been revealed.

Q20 MSQ

—random.choice() is used to select one movie title from the list.

—The unlock() function may reveal multiple occurrences of the guessed letter in a single attempt.

—The player's input is converted to uppercase before it is checked against the movie title.